

Lighting Control System Scenario

Interior decorating lighting designers have looked at the apps that are available for “smart bulbs” such as the Phillips Hue. Their complaints revolve around how the lights are grouped, how “scenes” are created, how colors are chosen, the lack of transition timing between scenes and the general look and feel of the interfaces.

The lighting designers have a set of objectives that they are looking achieve, but they are looking to your team to engineer the app appropriately.

The lights are multi-colored (RGB) and dimmable. The APIs supplied by the manufacturer for apps will do the heavy lifting of sending the correct codes to the lights for the color and brightness level. The app is the interface that allows the designer to choose the desired color and brightness.

Below are listed the major objectives:

- Each light (called the “load”) will need to be uniquely identified.
- Lights need to be grouped (from a single light to all of the lights) into “scenes” in such a way that either each individual light can be set to a specific color and brightness level or a set of lights can be set to the same color and brightness level.
- There needs to be a setting to allow for the transition time between scenes. This time is measured in seconds starting at 0 (the lights cannot change instantaneously, 0 just means no additional transition time).
- Scenes will need to be uniquely identifiable by name.
- There will need to be two default scenes. One where the all of the lights in the scene can be turned off and another where all of the lights are all on at full brightness and default (almost white) color.
- The design part of the app needs to be separated from the operational part of the app. Additionally, the design part of the app needs to have a method to prevent accidental scene design changes.